

# Extended continued fractions and energies of the anharmonic oscillators

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We describe the analytic solution to the Schrödinger eigenvalue problem for the class of the central potentials  $V(r) = \sum_{\delta \in Z} a_\delta r^\delta$ , where  $a_{-2} > -1/4$ ,  $a_{\max \delta} > 0$ ,  $Z$  is an arbitrary finite set of the integer or rational exponents,  $-2 \leq \delta_1 < \delta_2 < \dots < \delta_l$ , and the couplings  $a_\delta$  satisfy only one auxiliary (formal, "superconfinement") restriction of the type  $a_{\delta_i - 1} > 0$ . The formalism is based on an analysis of the asymptotic behavior of the explicit regular solution  $\psi(r)$  and issues in the formulation of the "secular" equation  $1/L_1(E) = 0$  which determines the binding energies. The main result is the rigorous construction of  $L_1(E)$  as a generalized ("extended") and convergent continued fraction. The method cannot be applied to the  $a_{\delta_i - 1} < 0$  cases—this disproves the closely related Hill-determinant approach as conjectured recently by Singh *et al.* for the simplest potentials with  $Z = \{-2, 2, 4, 6\}$  and  $Z = \{-2, -1, 1, 2\}$ .

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## I. INTRODUCTION

In the quantum field theory, the radial Schrödinger equation

$$-\frac{d^2}{dr^2} \psi(r) + \frac{l(l+1)}{r^2} \psi(r) + V(r)\psi(r) = E\psi(r) \quad (1.1)$$

$$l(l+1) = g_{-1} > -\frac{1}{4}$$

with the potential

$$V(r) = \sum_{j=1}^{2q+1} g_j r^{2j}, \quad g_{2q+1} = a^2 > 0, \quad (1.2)$$

appears in models with the simplest class of interactions.<sup>1</sup> It possesses the elementary solutions in some particular cases.<sup>2</sup> The more standard applications of Eq. (1.1) with the radial coordinate  $r \in (0, \infty)$  and centrifugal interpretation of  $g_{-1} = l(l+1)$ ,  $l = 0, 1, \dots$ , range from the perturbed harmonic oscillations of various systems<sup>3</sup> up to the structure of quarkonium.<sup>4</sup>

The knowledge of the particular solutions to Eq. (1.1)<sup>2</sup> is insufficient in most cases, and the nonnumerical construction of the complete solution represents a challenge to the mathematical physicists: The simple-minded perturbation expansions fail to give the convergent results.<sup>5</sup>

To overcome the methodical difficulties connected with the anharmonic equation (1.1), a number of alternative approaches have been developed recently—let us mention here just the moment recursions,<sup>6</sup>  $p-x$  symmetrization,<sup>7</sup> matrix continued fractions<sup>8</sup> and the continued-fraction method suggested for  $q=1$  by Singh *et al.*,<sup>9</sup> generalized to any  $q \geq 1$  in Ref. 10, and to an arbitrary fractionally anharmonic oscillator (FAO):

$$V(r) = \sum_{\delta \in Z} a_\delta r^\delta = \sum_{i=1}^l \gamma_i r^{-2 + m_i/n_i}, \quad (1.3)$$

$$0 \leq m_1/n_1 < \dots < m_l/n_l$$

in Ref. 11.

Our present paper has been inspired by the misleading Hill-determinant interpretation of the results in Refs. 9, 10, which was criticized by a few authors.<sup>12</sup> In brief, our inten-

is to clarify the situation and to prove in a rigorous way the following:

(A) For the potential (1.2), the secular equation of the Hill-determinant type may be written in the form

$$1/L_1(E) = 0, \quad (1.4)$$

where  $L_1(E)$  is an extended continued fraction<sup>13</sup> (ECF) to be defined below.

(B) For the potential (1.2), the roots of Eq. (1.4) coincide with the Schrödinger eigenvalues provided that the couplings satisfy the "superconfinement" restriction  $g_{2q} > 0$  or

$$g_{2q - m_i} > 0, \quad g_{2q - m_i + i} = 0, \quad i = 1, 2, \dots, m_0, \quad (1.5)$$

for some nonnegative integer  $m_0 \leq q-1$ .

(C) The validity of the statements (A) and (B) may simply be extended to the general FAO potential (1.3).

The material is organized as follows. In Sec. II, we summarize or reformulate some of the results of Refs. 10 and 11 and describe the analytic form of the general solution to the differential equation (1.1) with the FAO potential (1.3). In Sec. III, we present an analysis of the asymptotic structure of  $\psi(r)$  and complement it in Sec. IV by a thorough investigation of the other related formal questions. Finally, it is relatively straightforward to complete our discussion in Sec. V and to identify the convergent ECF quantity  $L_1(E)$  with the physical "Green's" function in all the SFAO ("superconfining" FAO) cases. In this way, Eq. (1.4) may be interpreted as an analytic sum of the (divergent, Brillouin-Wigner) perturbation expansion of the binding energy, or as a direct, though implicit, analog of the well-known harmonic-oscillator quantization requirement

$$E = \hbar\omega(2n + l + \frac{3}{2}).$$

## II. GENERAL SOLUTION TO THE DIFFERENTIAL EQUATION

By the direct insertion, we may easily verify that the explicit analytic solution of the radial Schrödinger equation

(1.1) with the generalized harmonic potential Eq. (1.2) has the closed form

$$\psi(r) = r^{\nu_{\pm}} \exp[-f(r)] \varphi(r), \quad \nu_{\pm} = \frac{1}{2} \pm (\frac{1}{4} + g_{-1})^{1/2}, \quad (2.1)$$

$$f(r) = \sum_{j=1}^{q+1} \frac{\beta_j r^{2j}}{2j}, \quad \varphi(r) = \sum_{n=0}^{\infty} h_{n+1} r^{2n},$$

where the Taylor coefficients have the form

$$h_{n+1} = h_1 \frac{\Gamma(\nu + \frac{1}{2}) \det Q(n)}{4^n n! \Gamma(n + \nu + \frac{1}{2})} \quad (2.2)$$

and the  $(q+2)$ -diagonal,  $(n \times n)$ -dimensional matrix  $Q(n)$  is defined by the prescription ( $E = -g_0$ )

$$\begin{aligned} Q(n)_{jj+1} &= B_j = -4j(j + \nu - \frac{1}{2}), \quad j = 1, 2, \dots, n-1, \\ Q(n)_{j+k,j} &= C_{j+k}^{(k)} = 4\beta_{k+1}(j-1 + \nu/2) + G_k, \\ j &= 1, 2, \dots, n-k, \\ G_k &= g_k - \sum_{i=1}^k \beta_i \beta_{k+1-i} + \beta_{k+1}(2k+1), \quad k = 0, 1, \dots, q. \end{aligned} \quad (2.3)$$

The function  $f(r)$  is closely related to the potential,  $V(r) = [\partial_r f(r)]^2 + O(r^{2q})$ ,  $r \gg 1$ , i.e.,

$$\begin{aligned} \beta_{q+1} &= a = g_{2q+1}^{1/2}, \\ \beta_k &= \frac{1}{2\alpha} \left( g_{q+k} - \sum_{j=k+1}^q \beta_j \beta_{k+q+1-j} \right), \\ k &= q, q-1, \dots, 1. \end{aligned} \quad (2.4)$$

When we introduce the new (barred) variables<sup>11</sup>

$$\bar{r} = r^p, \quad \bar{\psi}(\bar{r}) = r^{p-1/2} \psi(r), \quad 1 \leq p \leq 2q+2, \quad (2.5)$$

the radial Schrödinger equation (1.1) transforms into itself, with the barred centrifugal and energy parameters

$$\bar{g}_{-1} = (\frac{1}{4} + g_{-1})/p^2 - \frac{1}{4}, \quad \bar{E} = -g_{p-1}/p^2 \quad (2.6)$$

and with the new, broader class of potentials

$$\begin{aligned} \bar{V}(\bar{r}) &= \bar{g}_{-1} \bar{r}^{-2} + V_{(-)}(\bar{r}) + V_{(+)}(\bar{r}), \\ V_{(\pm)}(x) &= p^{-2} \sum_{m=1}^{M_{\pm}} g_{p-1 \pm m} x^{\pm 2m/p} \\ M_{+} &= 2q+2-p, \quad M_{-} = p-1 \end{aligned} \quad (2.7)$$

equivalent to the FAO forces (1.3) and containing Eq. (1.2) as a  $p=1$  special case, of course.

**Theorem 1:** For any FAO potential written in the form of Eq. (2.7), the regular solution  $\bar{\psi}(\bar{r})$  to the radial Schrödinger equation is given by Eq. (2.5), where  $\psi(r)$  is defined by Eqs. (2.1)–(2.4) and  $\nu = \nu_{+}$ . Similarly the irregular solution is obtained when  $\nu = \nu_{-}$ .

*Proof:* By insertion, we find that  $\bar{\psi}(\bar{r})$  is a solution. The regularity of the new equation in the origin is guaranteed since  $\bar{g}_{-1} > -\frac{1}{4}$  if and only if  $g_{-1} > -\frac{1}{4}$ , and an estimate  $\psi(r) \sim r^{\nu_{+}}$ ,  $r \ll 1$ , is equivalent to  $\bar{\psi}(\bar{r}) \sim \bar{r}^{\bar{\nu}_{+}}$ ,  $\bar{r} \ll 1$ , since  $\bar{\nu}_{\pm} = \frac{1}{2} + (\nu_{\pm} - \frac{1}{2})/p$ . QED

Let us add the following three remarks.

(i) For the given set  $Z_{\text{FAO}}$  of exponents  $\delta_i = n_i/m_i$ ,  $i$  is useful to determine the minimal value of  $q$  in practice. This reconstruction of Eq. (2.7) from  $V_{\text{FAO}}$  may be done in two steps. First, we divide all  $\delta$ 's in  $Z_{\text{FAO}}$  by 2, find their minimal common denominator  $M$  ( $\delta_i = 2N_i/M$ ), and put  $p = M \times T$ ,

TABLE I. Sample of the “complexities” of the simplest FAO one-term potentials  $V(r) = gr^{N/M}$ .

| $N \backslash M$ | 1 | 2 | 3  | 4 | 5  | 6 |
|------------------|---|---|----|---|----|---|
| −4               | 3 | 1 | 3  | 0 | 3  | 1 |
| −3               | 2 | 2 | 0  | 2 | 2  | 0 |
| −2               | 1 | 0 | 1  | 0 | —  | — |
| −1               | 0 | 0 | —  | — | —  | — |
| 1                | 2 | 0 | 4  | 2 | 6  | 1 |
| 2                | 4 | 2 | 6  | 0 | 8  | 4 |
| 3                | 6 | 1 | 2  | 4 | 10 | 0 |
| 4                | 8 | 4 | 10 | 2 | 12 | 6 |

where  $T$  should be a minimal positive integer. Second, we must guarantee that the energy term is not omitted from the Schrödinger equation and that the leading exponent  $2N_i/M = \max(\delta, 0)$  has the particular form  $2(2q+2-p)/p$ , i.e.,  $(N_i + M)T = 2q+2$ . Thus, for even  $N_i + M = 2k$  we choose  $T=1$  and put  $q=k-1$  while for odd  $N_i + M = 2k+1$  we must take  $T=2$  and  $q=2k$ . This procedure defines the minimal parameter  $q$  uniquely—a few examples are shown in Table I.

(ii) The choice of  $p=2q+2$  in Eq. (2.5) is a little bit exceptional since  $V(\infty) = V_{(-)}(\infty) = 0$  and our “spectral” restriction  $g_{2q+1} > 0$  eliminates in effect the continuous part of the spectrum (cf., e.g., the well-known  $q=0$  harmonic oscillator→Coulomb-potential transformation). At the same time, when  $g_{2q-m} = 0$ ,  $m=0, 1, 2, \dots, m_0-1$ , and  $g_{2q-m_0} \neq 0$ , the “subdominant anharmonicity”  $m_0$  is to be added to the classification of  $V = V_{\text{FAO}}$  and  $\psi$  by the “complexity”  $q$  and the “fractionality”  $p$ .

(iii) The choice of  $p=1$  with  $g_{-1}=0$  seems to be also exceptional—it admits the ( $r \leftrightarrow -r$  symmetric) one-dimensional interpretation of Eq. (1.1) and  $\psi(r)$ , with  $r \in (-\infty, \infty)$  and negative and positive parity for  $\nu = \nu_{+}$  and  $\nu = \nu_{-}$ , respectively. For  $q=0$ , the series  $\varphi(r)$  in Eq. (2.1) coincides with the well-known confluent hypergeometric function, while for  $q=1$ , it reproduces precisely the sextic-oscillator functions of Ref. 9.

### III. ASYMPTOTIC BEHAVIOR OF THE REGULAR SOLUTIONS

In analogy with the Singh's papers,<sup>9</sup> we may achieve in principle even the termination of the infinite series  $\varphi(r)$  in Eq. (2.1) for some potentials.<sup>2</sup> For this purpose, it is sufficient to require the validity of the  $q+1$  independent algebraic conditions (determinantal constraints)

$$h_{N+i} = 0, \quad i = 1, 2, \dots, q+1. \quad (3.1)$$

They fix the energy plus  $q$  couplings—the thorough discussion of this peculiarity may be found in Refs. 2 and shows that we get at most two polynomial bound states when  $q > 3$ . The simplest harmonic-oscillator and Coulombic  $q=0$  forces appear to be the only exceptions giving the complete set of the terminating solutions (Laguerre polynomials) accompanied by the explicit formula (i.e.,  $h_{N+1} = 0$ ) for the energy.

Reverting the preceding argument, we may conclude that the normalizable  $q \geq 1$  bound-state solutions should be

represented by an infinite series in general. In accord with the standard textbooks,<sup>14</sup> this series is convergent and its absolute value grows in the asymptotic region, provided that the energy  $E$  does not coincide with the physical eigenvalues  $E_n$ . At  $E = E_n$  the value of  $\psi(r_0)$  changes sign in the limit  $r_0 \rightarrow \infty$ —this may be used in the numerical determination of the discrete spectrum.

In the present paper, the structure of  $\psi(r)$  and its asymptotic behavior in the vicinity of  $E_n$  will be clarified by the decomposition of the regular solution Eq. (2.1) into the  $q + 1$  partial infinite summations

$$\begin{aligned}\psi(r) &= \sum_{j=0}^q \chi_j(r), \\ \chi_j(r) &= r^{\nu} e^{-f(r)} \sum_{k=0}^{\infty} h_{n_j(k)+1} r^{2n_j(k)}, \\ n_j(k) &= k(q+1) + j.\end{aligned}\quad (3.2)$$

Let us emphasize that the value of fractionality  $p$  is irrelevant in this context, of course—it concerns just the interpretation of one of the constants ( $-g_{p-1}/p^2$ ) as energy.

**Proposition 1:** The necessary and sufficient condition for the FAO regular wavefunction  $\psi(r) = \chi_0(r) + \dots + \chi_q(r)$  to be normalizable is either its termination or an asymptotic cancellation of its  $q + 1$  growing exponential components  $\chi_j(r) \sim \exp[f(r)]$ ,  $r \gg 1$ .

**Proof:** We introduce an auxiliary sequence  $L_n^{(N)}$  defined by the recurrences

$$L_k^{(N)} = -B_{k-1} / \left[ C_k^{(0)} + \sum_{j=1}^q \left( \prod_{m=0}^{j-1} L_{k+m+1}^{(N)} \right) C_{k+j}^{(j)} \right], \quad (3.3)$$

$L_{N+1} = L_{N+2} = \dots = 0$ ,  $k = N, N-1, \dots, 1$ ,  $B_0 \equiv 1$ , abbreviate the combinations

$$U_{m+k}^{(m)} = C_{m+k}^{(m)} + L_{m+k+1}^{(N)} U_{m+k+1}^{(m+1)}, \quad 1 \leq m \leq q-1, \quad (3.4)$$

$U_{q+k}^{(q)} = C_{q+k}^{(q)}$ ,  $U_{m+k}^{(m)} = 0$ ,  $m > q$ ,  $k = 1, 2, \dots, N-m$ , and define formally the product decomposition of the matrix Eq. (2.3)

$$Q(N) = -X(N)Y(N)Z(N) \quad (3.5)$$

with the nonzero elements

$$\begin{aligned}X(N)_{kk} &= 1, \quad Z(N)_{kk} = B_{k-1}, \\ Y(N)_{kk} &= 1/L_k^{(N)}, \quad 1 \leq k \leq N, \\ X(N)_{k,k+1} &= -L_{k+1}^{(N)}, \quad Z(N)_{k,m+k} = -L_{k+m}^{(N)} U_{m+k}^{(m)},\end{aligned}\quad (3.6)$$

$1 \leq m \leq \min(q, N-k)$ ,  $1 \leq k \leq N-1$ .

For the Taylor coefficients (2.2) we then get

$$h_{N+1} = h_1 / L_1^{(N)} L_2^{(N)} \dots L_N^{(N)}. \quad (3.7)$$

As a rule, the singularities of the type  $L_k^{(N)} = \infty$  are to be understood as limits of the regular cases.

In the  $n \gg 1$  asymptotic region we have  $C_n^{(j)} = 4n\beta_{j+1}$ ,  $j = 0, 1, \dots, q$ ,  $B_n = -4n^2$  and

$$L_n^{(N)} = n / \left( \beta_1 + \sum_{j=1}^q \beta_{j+1} \prod_{m=1}^j L_{n+m}^{(N)} \right) \quad (3.8)$$

within the  $1 + O(1/n)$  error bounds. As a consequence we obtain the first few elements of the sequence  $L_n^{(N)}$  in the form

$$\begin{aligned}L_{N-(q+1)k}^{(N)} &= N / \beta_1^{(k)}, \quad L_{N-(q+1)k-j}^{(N)} = \beta_j^{(k)} / \beta_{j+1}^{(k)}, \\ j &= 1, 2, \dots, q,\end{aligned}\quad (3.9)$$

where  $\beta_j^{(0)} = \beta_j$  and

$$\beta_j^{(k+1)} = \beta_j + \frac{1}{\alpha} \sum_{m=1}^{q+1-j} \beta_{q+1-m}^{(k)} \beta_{j+m} \quad (3.10)$$

$$j = 1, 2, \dots, q, \quad \beta_{q+1}^{(k+1)} = \alpha, \quad k = 0, 1, \dots$$

The products  $L_{N-(q+1)k}^{(N)} \times L_{N-(q+1)k-1}^{(N)} \times \dots \times L_{N-(q+1)k-q}^{(N)} = [N + O(1)]/\alpha$  may therefore be inserted into the definition (3.7) and the coefficients  $h_{N+1}$  and their  $N$  dependence may be given the form

$$h_{N+1} = [(\alpha\Delta)^{N\Delta} / \Gamma(N\Delta + 1)] b_{N+1}, \quad \Delta = 1/(q+1). \quad (3.11)$$

In the asymptotic region  $N \gg 1$ , the new coefficients  $b_{N+1} \sim b_{N-q}$  exhibit at most the  $(q+1)$ -periodic oscillations so that the  $q+1$  infinite partial summations in Eq. (3.2) exhibit the same exponential behavior  $\exp[2f(r)]$  for  $r \gg 1$ . QED

Let us note that for  $q = 0$  the cancellation is impossible ( $\psi \equiv \chi_0$ ) while, for  $q \geq 1$ , the termination of some particular states (with  $b_N = 0$ ,  $N \gg 1$ , and  $b_{N+1}^{(N)} = 0$ ; cf. Ref. 2) belongs in fact to the same exceptional phenomena as a nonexistence of our fundamental decomposition (3.5) and will not be discussed here in any detail. Hence the only way how to normalize the harmonic oscillator is the textbook termination of the infinite series  $\varphi(r)$  while, for the anharmonic oscillators, the universal mechanism of satisfying the physical boundary conditions is represented by the asymptotic cancellation of  $\chi$ 's. For  $q = 1$ , this cancellation was shown in Refs. 9 and 12 to follow from the convergence of the analytic continued fractions  $L_n = L_n^{(\infty)} = \lim_{N \rightarrow \infty} L_n^{(N)}$ . Similar limits (ECF quantities<sup>13</sup>) may be defined for  $q > 1$  as well—their relevant properties will be described in the next section.

#### IV. EXTENDED CONTINUED FRACTIONS AND THEIR CONVERGENCE

For large  $n = O(N) \gg 1$ , the finite approximants  $L_n^{(N)}$  depend only on the difference  $N - n$  and oscillate—this is a consequence of Eqs. (3.8)–(3.10). Provided that the ECF limit  $L_n^{(\infty)}$  exists, these  $(q+1)$ -periodic oscillations must be suppressed, i.e.,  $L_n \doteq L_{n+1} = \dots \doteq L_{n+q} \doteq P_n$  for  $O(1) \ll n \ll O(N)$ . The  $P_n$ 's [stationary points of the  $q$ -to-one mapping Eq. (3.3)] have to satisfy the algebraic equation

$$P_n = n / (\beta_1 + \beta_2 P_n + \dots + \beta_{q+1} P_n^q) \quad (4.1)$$

and may be given explicitly by the asymptotic expansion of the type

$$P_n = P_n^{(\epsilon)} = \epsilon \left( \frac{n}{\alpha} \right)^\Delta - \frac{\beta_q \Delta}{\alpha} + O\left( \frac{1}{n} \Delta \right), \quad (4.2)$$

$$\Delta = \frac{1}{(q+1)}, \quad \epsilon = \epsilon_m = \exp(2\pi i m \Delta), \quad m = 0, 1, \dots, q.$$

As a consequence, we arrive at an asymptotic estimate

$$L_n^{(N)} = P_n^{(\epsilon)} \left( 1 + O\left(\frac{1}{n}\right) \right), \quad N \gg n, \quad (4.3)$$

which enables us to accelerate the convergence: We put

$$L_n = P_n^{(\epsilon)} + R_n \quad (4.4)$$

and redefine the initialization,  $R_{N+1}^{(N)} = 0$  in the limit  $N \rightarrow \infty$ .

**Proposition 2:** Provided that  $0 \leq m_0 \leq q-1$  and

$$\beta_{q-m} = 0, \quad m = 0, 1, \dots, m_0 - 1, \quad \beta_{q-m_0} \epsilon^{q-m_0} > 0, \quad (4.5)$$

the finite ECF approximants  $R_n^{(N)} = L_n - P_n^{(\epsilon)}$  defined by the recurrences (3.3) and initialization  $R_{N+1}^{(N)} = 0$  are convergent and determine the ECF function  $R_n^{(\infty)}$ .

**Proof:** The new nonlinear ECF recurrences for  $R_n^{(N)}$  follow directly from an insertion of the definition (4.4) into the old recurrences (3.3). They have the asymptotic form

$$R_n^{(N)} = - \frac{P_n^2 S_n(R_{n+1}^{(N)}, \dots, R_{n+q}^{(N)}) [1 + O(1/n)]}{n + P_n S_n(R_{n+1}^{(N)}, \dots, R_{n+q}^{(N)})}, \quad (4.6)$$

$$\begin{aligned} S_n(R_{n+1}, \dots, R_{n+q}) &= \beta_2(L_{n+1} - P_n) + \dots + \beta_{q+1}(L_{n+1} \dots L_{n+q} - P_n^q) \\ &= \sum_{s=1}^q \beta_{s+1} P_n^{s-1} \sum_{k=1}^s R_{n+k} \left[ 1 + \sum_{l=1}^{k-1} \sum_{t_1=1}^l \dots \right. \\ &\quad \left. \dots \sum_{t_{l-1}=1}^{t_l-1} \prod_{m=1}^l \frac{R_{n+l+t_m-m}}{P_n} + O\left(\frac{1}{n}\right) \right], \end{aligned}$$

with the zero (i.e.,  $O(1/n^{1-\Delta})$ ) stationary point.

Having in mind the error bounds, we may specify the vicinity of "zero" in the form

$$R_n^{(N)} = O(n^{(1-\Delta)/2}/n^{1-\Delta}) \gg O(1/n^{1-\Delta}). \quad (4.7)$$

In the asymptotic region, this permits the linearization of the ECF recurrences (4.6),

$$\begin{aligned} R_n &= -(t_1 R_{n+1} + \dots + t_q R_{n+q}) + O(1/n^{1-\Delta}) \\ t_k &= 1 - (\beta_k P_n^k + \dots + \beta_1 P_n)/n + O(1/n), \\ k &= 1, 2, \dots, q. \end{aligned} \quad (4.8)$$

Let us consider any initialization

$$R_{N+i}^{(N)} = \rho_i = O(1/N^{(1-\Delta)/2})$$

and rewrite the linearized recurrences (4.8) in the matrix form

$$\begin{pmatrix} 1 & 0 & \dots & \dots \\ t_1 & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots \\ t_{q-1} & \dots & 1 & 0 & \dots \\ t_q & t_{q-1} & \dots & \dots & \dots \\ 0 & t_q & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} R_N \\ R_{N-1} \\ \dots \\ R_{N-q+1} \\ R_{N-q} \\ R_{N-q-1} \\ \dots \end{pmatrix} = \begin{pmatrix} \tilde{\rho}_1 \\ \tilde{\rho}_2 \\ \dots \\ \tilde{\rho}_q \\ 0 \\ 0 \\ \dots \end{pmatrix}, \quad (4.9)$$

where  $\tilde{\rho}_i, i = 1, 2, \dots, q$ , are some simple linear  $O(1/N^{(1-\Delta)/2})$  combinations of  $\rho_i$ 's. Using the decomposition of the type

$$\begin{pmatrix} 1 & 0 & \dots \\ t_1 & 1 & 0 & \dots \\ t_2 & t_1 & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots \\ -a_1 & 1 & 0 & \dots \\ 0 & -a_1 & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \times \dots \times \begin{pmatrix} 1 & 0 & \dots \\ -a_q & 1 & 0 & \dots \\ 0 & -a_q & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (4.10)$$

where  $a_1, \dots, a_q$  are complex numbers in general, we may invert easily the left-hand-side matrix in Eq. (4.9) and get

$$\begin{pmatrix} R_N \\ R_{N-1} \\ R_{N-2} \\ \dots \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots \\ a_q & 1 & 0 & \dots \\ a_q^2 & a_q & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \times \dots \times \begin{pmatrix} 1 & 0 & \dots \\ a_1 & 1 & 0 & \dots \\ a_1^2 & a_1 & 1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \times \begin{pmatrix} \tilde{\rho}_1 \\ \tilde{\rho}_2 \\ \tilde{\rho}_3 \\ \dots \end{pmatrix}. \quad (4.11)$$

In the next step, we may prove by induction that the right-hand-side product  $M_{m,n} = M_{m+k,n+k}, k = 1, 2, \dots$ , of matrices in Eq. (4.11) is zero for  $n > m$  and its nonzero elements may be written in the compact form

$$\begin{aligned} M_{m+k,m} &= \frac{a_1^{q+k}}{(a_1 - a_2)(a_1 - a_3) \dots (a_1 - a_q)} \\ &\quad + \frac{a_2^{q+k}}{(a_2 - a_1)(a_2 - a_3) \dots (a_2 - a_q)} + \dots \\ &\quad + \frac{a_q^{q+k}}{(a_q - a_1)(a_q - a_2) \dots (a_q - a_{q-1})}, \\ m &= 1, 2, \dots, \quad k = 0, 1, \dots. \end{aligned} \quad (4.12)$$

Hence, the  $k$  dependence of  $R_{N-k}$  is given by the powers of  $a_i^k$ . In the final step of the proof, it is sufficient to show that  $\|a_i\| < 1, i = 1, 2, \dots, q$ .

By straightforward algebra, we get  $t_1 = -a_1 - a_2 - \dots - a_q, \dots, t_q = (-1)^q a_1 a_2 \dots a_q$  in Eq. (4.10) so that  $a_i$ 's are equal to the roots of the algebraic equation  $x^q + t_1 x^{q-1} + \dots + t_q = 0$  which may be written in the form

$$\frac{1 - x^{q+1}}{1 - x} - \frac{\beta_1 P_n}{n} \frac{1 - x^q}{1 - x} - \dots - \frac{\beta_q P_n^q}{n} \frac{1 - x}{1 - x} = O\left(\frac{1}{n}\right), \quad x \neq 1. \quad (4.13)$$

Now, assuming that  $n$  is sufficiently large and Eq. (4.5) is satisfied, we get

$$1 - x^{q+1} = \frac{\beta_{q-m_0} P_n^{q-m_0}}{n} (1 - x^{m_0+1}) [1 + O(1/n^{\Delta})]. \quad (4.14)$$

Since the right-hand-side of Eq. (4.14) may be made arbitrarily small by the choice of  $n \gg 1$ , we may write

$$a_l = \epsilon_l(1 + \eta), \quad \epsilon_l = e^{2\pi i \Delta l}, \\ 1 \leq l \leq q, \quad \|\eta\| \ll 1. \quad (4.15)$$

Inserting this expression into Eq. (4.14), we obtain

$$\eta = \beta_{q-m_0} P_n^{q-m_0} (\epsilon^{m_0+1} - 1) \Delta / n + O(\eta^2). \quad (4.16)$$

In accord with our assumption (4.5), the real part of  $\eta$  is always negative so that  $\text{Re}(1 + \eta) < 1$  and  $\|a_l\| < 1$  for  $n < \infty$ . This completes the proof since, in accord with Eq. (4.15), the denominators in Eq. (4.12) are all nonzero, and  $R_{N-k} = O(\max \|a_i^k\|) = O(\exp(-\text{const} k/n^d))$  may be made arbitrarily small within the  $N^d \ll k \ll N$  asymptotic region. QED

(i) On the particular  $q = 1$  example, the content of Proposition 2 may be illustrated very simply—the ECF degenerates to an ordinary continued fraction,<sup>15</sup> the choice between the two roots of the quadratic Eq. (4.1) is given by the rule  $\epsilon = \text{sgn} \beta_q$  and both the initializations  $L_{N+1}^{(N)} = 0$  and  $R_{N+1}^{(N)} = 0$  are equivalent so that the transition  $L_n \rightarrow R_n$  [Eq. (4.4)] is merely an acceleration of convergence.

(ii) For  $q > 1$ , the values of  $P_n^{(\epsilon)}$  may be complex, causing a divergence of  $L_n^{(\infty)}$ , which need not necessarily be accompanied by the divergence of  $R_n^{(\infty)}$  *a priori*. The exact proofs of equivalence would be extremely complicated even for the real  $\epsilon$ 's ( $\epsilon = \pm 1$ )—an example of a  $q = 2$ —analog of Proposition 2 for  $L_n^{(\infty)}$ 's may be found in Ref. 10. Fortunately, the present weaker convergence of  $R_n^{(\infty)}$  is fully sufficient in the context of the next section.

## V. ENERGIES OF THE BOUND STATES

The determinantal equation (2.2) for  $h_{n+1}$  is less suitable in the asymptotic region—let us switch back to the original recurrences. They may be given the form of the infinite-dimensional matrix equation

$$Q(\infty) \begin{pmatrix} h_1 \\ h_2 \\ \dots \end{pmatrix} = 0. \quad (5.1)$$

Assuming that the product decomposition (3.5) of  $Q(N)$  exists also in the limit  $N \rightarrow \infty$ , we obtain the fully equivalent form of Eq. (5.1):

$$Y(\infty) Z(\infty) \begin{pmatrix} h_1 \\ h_2 \\ \dots \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \dots \end{pmatrix}, \quad (5.2)$$

where the right-hand-side vector is defined in terms of the convergent ECF's by the formulas  $X(\infty)\omega = 0$ , i.e.,

$$\omega_k = h_1 \prod_{j=1}^k L_j^{(\infty)}, \quad k = 1, 2, \dots, \quad (5.3)$$

and it can, of course, be nonzero in the  $N = \infty$  case. Hence we do not need to require *a priori* the zero Hill determinant  $\det[Q(\infty)] = 0$ , contrary to the intuitive expectations of Ref. 9, where Eq. (5.1) was misinterpreted as an eigenvalue problem.

The validity of the  $g_2 > 0$  part of the Singh's results is almost surprising—the physical energies coincide indeed with the zeros of the Hill determinant.<sup>12</sup> Let us show that this is a peculiarity of all the SFAO potentials [= FAO's

restricted by the “superconfinement” or “subdominant positivity” requirement (1.5)].

**Theorem 2:** When we put  $\epsilon = 1$  in the ECF definition (4.3), the roots  $E_n$  of the “secular” equation

$$1/R_1^{(\infty)}(E) = 0 \quad (5.4)$$

determine the physical SFAO bound-state energies.

*Proof:* First, we have to prove the convergence of the ECF quantities  $R_n^{(\infty)}$ . This is an easy task because Eq. (1.5) implies Eq. (4.5), and we may apply Proposition 2. Next, we may factor out all the dominant  $n$ -dependence in the recurrences Eq. (5.1). In detail, estimates of Eq. (4.3) and

$$L_n U_n^{(q-k)} = 4n\alpha P_n^{k+1} + 4n\beta_q P_n^k (1 - k\Delta) + \dots, \\ k = 0, 1, \dots, q-1 \quad (5.5)$$

make it possible for us to rewrite Eq. (5.2), i.e., the Taylor recurrences

$$h_k B_{k-1} / L_k^{(\infty)} = \omega_k + \sum_{i=1}^{\min(q, k-1)} U_k^{(i)} h_{k-i}, \quad k = 1, 2, \dots, \quad (5.6)$$

in the  $k \gg 1$  asymptotic form. Indeed, using Eq. (3.11) and the Stirling formula in the form

$$[(k/\Delta)!]^\Delta = k! \Delta^{-k-\Delta/2} (2\pi k)^{(\Delta-1)/2} [1 + O(1/k)], \quad (5.7)$$

we obtain the relation

$$\sum_{j=0}^q \epsilon^j b_{N+j} = \frac{\text{const}}{L_1^{(\infty)}} \left[ 1 + O\left(\frac{1}{N^\Delta}\right) \right], \quad (5.8)$$

reflecting the contribution of  $\omega_N$ 's, the size of which is comparable with the  $b_N$ 's.

Finally, the  $q+1$   $b$ 's may be considered almost constant in the  $N \gg 1$  asymptotic region,  $b_{N+j} \sim b_j$ . Hence, in accord with Eq. (5.8) and for the particular value  $\epsilon = 1$ , the sign of the superposition of  $\chi_j$ 's for  $r \gg 1$  is determined by the sign of  $L_1(E)$ . This sign changes exactly at the root  $E_n$  of Eq. (5.4) since the poles of  $L_1(E)$  and  $R_1(E)$  coincide. With respect to the standard oscillation theorems,<sup>14</sup> a new node appears in  $\psi$  whenever  $E$  crosses  $E_n$ —the value  $E_n$  coincides precisely with the SFAO bound-state energy. QED

(i) The proof given above is the main result of our paper. The Theorem 2 may be interpreted as a rigorous foundation of the Hill-determinant interpretation<sup>9</sup> of the eigenvalue problem with the following strong warning: The physical energies  $E_n$  in Ref. 9 coincide with the Hill-determinant zeros by mere chance—for the non-SFAO potentials with  $g_{2q-m_0} < 0$ , the Singh-type interpretation of  $\det[E - Q(\infty)]$  is completely misleading and gives the unphysical energies.<sup>12</sup>

(ii) Indeed, for  $\epsilon \neq 1$ , there is no relation between the root of Eqs. (5.4) and (1.4) and the asymptotic behavior of  $\psi$ . The superposition of  $b_j$ 's or  $\chi_j$ 's is in no way related to Eq. (5.8): For  $\epsilon = -1$  and  $q = 1$ , the asymptotic cancellation of  $\chi_0$  and  $\chi_1$  is even minimal<sup>12</sup> at  $E = E_n$ —the asymptotically decreasing exponential component of  $\psi(r)$  seems to be suppressed. Thus, the choice of  $\epsilon = +1$  in Theorem 2 is unique, and we cannot remove the “superconfinement” restriction within the present ECF framework.

(iii) The main merit of our “secular” Eqs. (5.4) or (1.4) is their analytic and compact structure. Nevertheless, in con-

trast to the original expectations,<sup>9</sup> the purely numerical exploitation of Eq. (5.4) is also possible in principle.

From this numerical point of view, the transition  $L_n^{(N)} \rightarrow R_n^{(N)}$  is extremely useful. When we consider the illustrative example  $\beta_i = 1$  with  $L_{N+1}^{(N)} = 0$  and

$$L_{N-(q+1)k}^{(N)} = \frac{q!k!}{(q+k)!} N, \\ L_{N-(q+1)k-j}^{(N)} = 1 + \frac{k}{q+1-j}, \quad (5.9) \\ j = 1, 2, \dots, q, \quad k = 0, 1, \dots$$

[cf. Eq. (3.9)], we see that the use of  $R_n^{(N)}$ 's eliminates the large  $(q+1)$ -periodic oscillations  $L_n^{(N)} - L_{n+1}^{(N)} = O(n)$  which survive in the asymptotic region for at least as many as  $O(N^4)$  iterations of the old recurrences, Eq. (3.3).

(iv) Of course, the smaller oscillations are present in  $R_n^{(N)}$ 's as well. Fortunately, they decrease for the higher cutoffs  $N$ . Moreover, they may systematically be suppressed either by the next ( $k$ th) subtraction of the form

$$R_n[k+1] = R_n[k] - P_n[k+1], \\ R_n[0] = L_n^{(N)}, \quad P_n[1] = P_n^{(1)}, \quad (5.10) \\ R_n[1] = R_n^{(N)}, \dots,$$

or by an averaging over the  $q+1$  neighboring cutoffs  $N = N_0 + i, i = 1, 2, \dots, q+1$ . The efficiency of the latter type of "smoothing" was found empirically in a somewhat related context in Ref. 6. When combined with the subtractions (5.10), it was tested in Ref. 10 for  $q = 2$  and proved to be comparable even with the specialized methods.

## VI. CONCLUDING REMARKS

We have proved that Eq. (5.4) defines the energies of the anharmonic oscillators which belong to the SFAO class, i.e., which are restricted by Eq. (1.5). We have also proved that the ECF representation of the underlying "Green's" function  $R_1^{(\infty)}(E)$  is convergent. In the conclusion we would like to add the following remarks:

(i) Rather surprisingly, the acceleration of convergence (quasiequivalence  $L_n \rightarrow R_n \rightarrow R_n[2] \rightarrow \dots$ ) makes the analytic ECF formalism well suited even for the numerical computations.

(ii) The immanent "superconfinement" restriction resembles the similar property of the JWKB method<sup>16</sup> and cannot be removed without the deep modifications of the ECF method. The possibility to revert the ECF recurrences is under current investigation at present.

(iii) The partial (SFAO) coincidence of the present secular equation with its Hill-determinant precursors<sup>9,10</sup> may be characterized as a lucky chance and attributed to the very special type of the Schrödinger boundary conditions imposed on  $\psi(r)$  at  $r \rightarrow \infty$ .

(iv) Contrary to the *a priori* expectations based on the analogy with the  $q = 0$  harmonic oscillator, the essential technicality leading to the successful completion of the standard power-series method is not only the factorization of the wavefunction  $\psi = \exp(-\text{polynomial}) \times \text{power series}$ , but also the ECF shortening of the Taylor recurrences to their final  $(q+1)$ -term form. The SFAO restriction (or the ECF convergence from the more formal point of view) is in fact just the condition of stability of the underlying decomposition  $Q(\infty) = X(\infty)Y(\infty)Z(\infty)$ .

(v) The exceptional (terminating) solutions<sup>2</sup> which correspond to the singular cases in our formulas ( $L_k \rightarrow \infty$  etc.) are to be interpreted as limits of the neighboring fully regular cases. This is compatible with the spirit of the classical theory of the analytic continued fractions.<sup>15</sup>

(vi) The classification of the FAO and SFAO forces by means of the "complexity"  $q$ , "fractionality"  $p$ , and "subdominant anharmonicity"  $m_0$  is an interesting byproduct of our considerations. It leads to an unusual partial ordering  $q = 0, q = 1, \dots$  of some standard anharmonic models.

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